

TECHNOLOGY TRANSFER AND ... INTELLECTUAL PROPERTY RIGHTS IN DEVELOPING COUNTRIES

Walter G. Park, American University
Douglas C. Lippoldt, OECD

February 2010

ISSUES

- ◉ How IPRs affect inward technology transfer
 - Particularly technology-intensive goods, services, and capital
 - ... beyond sales & distribution, low-wage production
- ◉ Effect of inward technology transfer on local innovation
 - Foreign technology as an input into local innovation
- ◉ Effect on capacity for outward technology transfer
 - Capacity of local firms to engage in outward technology transfer

BACKGROUND

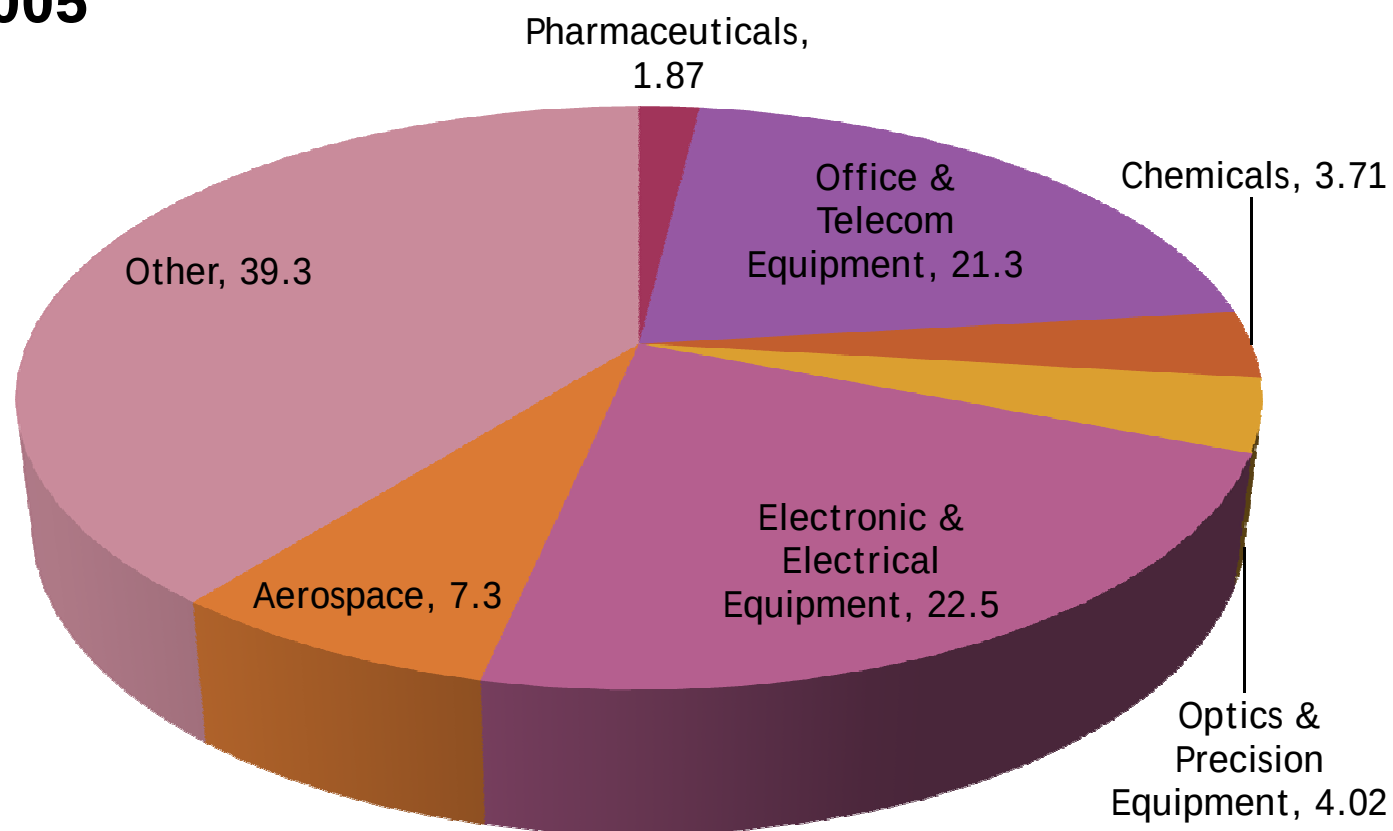
- ◉ Article 66.2 of TRIPS Agreement
 - Obligation of Developed Countries to assist in technology transfer to Least Developed.
- ◉ Modes of technology transfer
 - Foreign Direct Investment (FDI), Trade (Merchandise & Services), and International Licensing
- ◉ Trends Across Country Groups
 - Developed, Developing, and Least Developed (United Nations Classification)

TRENDS IN TECH. TRANSFER

Country Group	Mode	Mean 2005 (\$billions)	% Δ since 1995
Developed	Inward FDI Stock	\$244.9	169.8%
	Merchandise Imports	\$236.2	56.2%
	Service Imports	\$58.4	53.4%
Developing	Inward FDI Stock	\$35.8	234.3%
	Merchandise Imports	\$47.3	91.4%
	Service Imports	\$9.2	70.4%
Least Developed	Inward FDI Stock	\$2.05	262.8%
	Merchandise Imports	\$2.2	103.8%
	Service Imports	\$0.92	91.1%

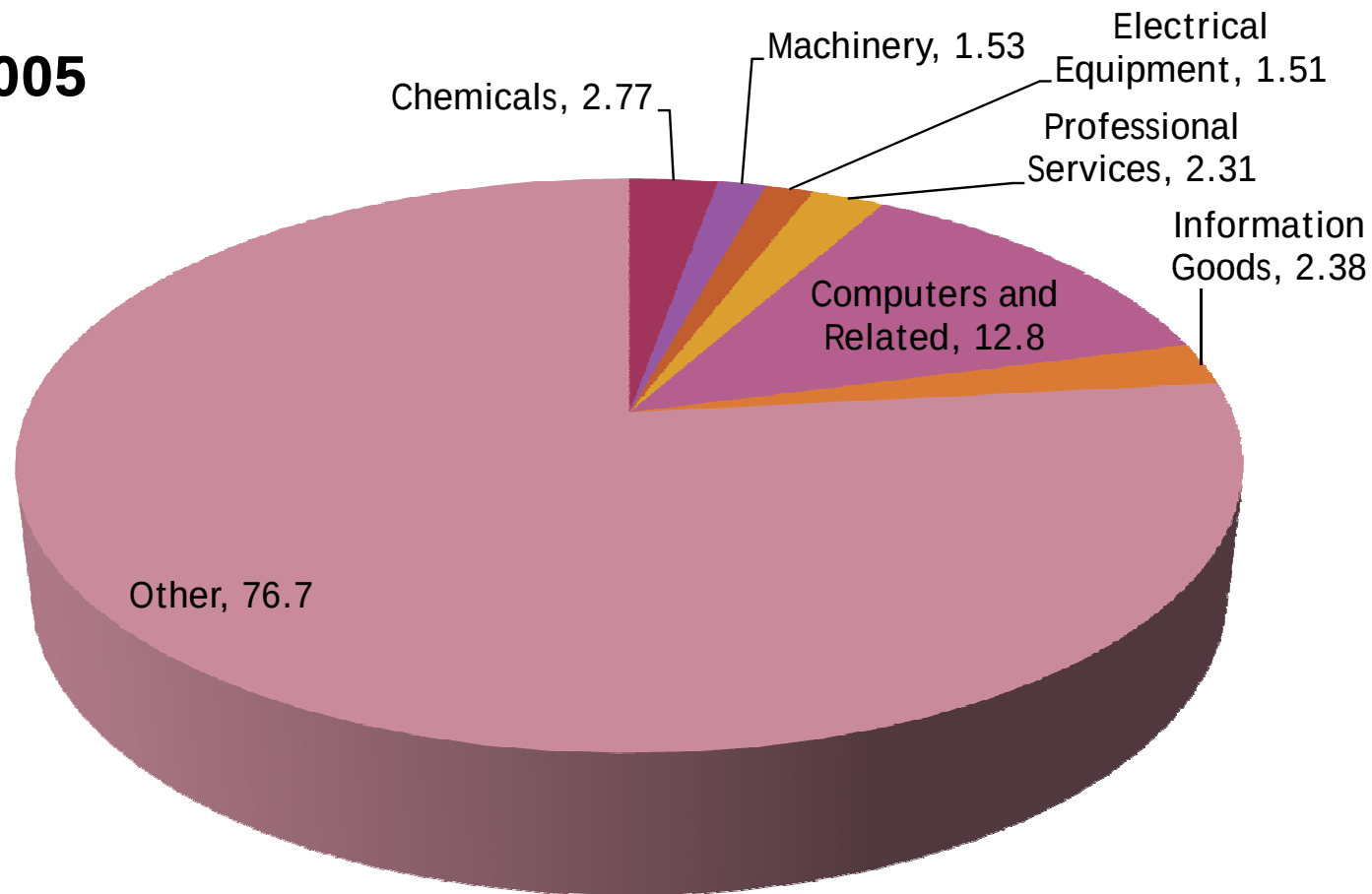
% COMPOSITION OF MERCHANDISE IMPORTS OF DEVELOPING & LEAST DEVELOPED COUNTRIES

2005



% COMPOSITION OF ASSETS OF FOREIGN AFFILIATES OF U.S. MNCs IN DEVELOPING & LEAST DEVELOPED COUNTRIES

2005



CONCEPTUAL FRAMEWORK

- ◉ Global Perspectives
 - North-South
- ◉ Trade, FDI, and Licensing as a “vehicle” for technology diffusion
- ◉ IPRs and Trade, FDI, Licensing
 - Market Expansion vs. Market Power
 - ◉ *Role of Imitative Capacity*
 - Ownership, Location, and Internalization (OLI)
 - Volume & Composition of Technology Transfers

PREVIOUS EVIDENCE

- ◉ Mansfield (1994), Lee and Mansfield (1996)
- ◉ Fink and Primo Braga (1998, 1999)
- ◉ Maskus (1998, 2004), Maskus et al. (2005)
- ◉ Smith (1999, 2001)
- ◉ Mayer and Pfister (2001)
- ◉ Javorcik (2004)
- ◉ Nunnenkamp and Spatz (2004)
- ◉ Park and Lippoldt (2005)
- ◉ Branstetter et al. (2006, 2007)
- ◉ Nicholson (2007)

GAPS IN PREVIOUS WORK

◉ Dated

- Evidence before TRIPS Agreement (1995)
- Location vs. Volume of transfers

◉ Unresolved Issues

- Have FDI to developing countries been technology-intensive?
- Vintage of technology?
- Controls for other institutional factors

TASKS

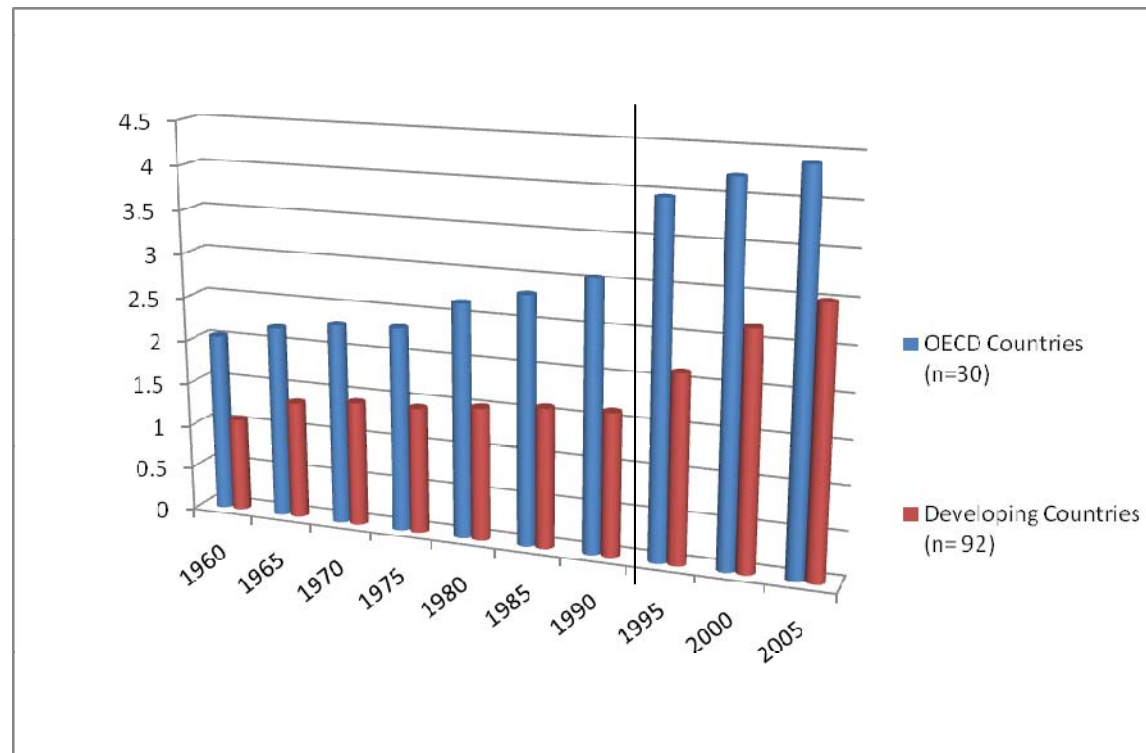
- ◉ Measures of intellectual property rights
 - Patent Protection
 - Copyrights
 - Trademarks
 - IPR Survey (perceptions of enforcement adequacy)
- ◉ FDI, Trade by sector breakdown
 - Do IPRs stimulate technologically-intensive technology transfers?
- ◉ Other institutional controls
 - Property rights in general, legal effectiveness, 'Doing Business' index, governance, trade policy

GLOBAL IPR DEVELOPMENTS

Country Group	Index of Patent Rights 0 - 5 (%Δ since 1995)	Index of Copyrights 0 - 1 (%Δ since 1995)	Index of Trade-Mark Rights 0 - 1 (%Δ since 1995)	Executive Opinion Survey (IPR) 1- 7
Developed	4.4	0.80	0.70	5.5
	(10%)	(14.3%)	(16.7%)	
Developing	3.3	0.60	0.60	3.5
	(37.5%)	(20%)	(20%)	
Least Developed	2.4	0.42	0.36	2.7
	(26.3%)	(0%)	(33.3%)	

Correlations	Patent	Copy	T Mark	IPR Survey
Patent	1			
Copy	0.53	1		
T Mark	0.57	0.58	1	
IPR Survey	0.62	0.36	0.37	1

EVOLUTION OF THE PATENT RIGHTS INDEX, 1960-2005



- The vertical bar indicates the advent of the TRIPS Agreement.

EMPIRICAL ANALYSIS: PANEL DATA

◉ Dataset

- 122 countries
- 1990 - 2005

◉ Dependent Variables of Interest

- Inward FDI (acquisition, expansion of facilities)
- Merchandise Trade (source of capital goods)
- Services Trade (FDI is conduit for services)

◉ Perspectives

- Different types of IPRs
- Different groups of countries
- Different industries

RESULTS

- ◉ Patent rights important to FDI, trade
 - Copyright, trademark not statistically significant
 - Enforcement adequacy also important

- ◉ Across country groups
 - Quantitative impact larger in Developed countries (role of complementary factors)
 - IP Statutes important in Least Developed countries (more than perceptions of enforcement)
 - IP has 'market power effects' in smaller markets
 - Developing country group is relatively most heterogeneous

RESULTS ...

- ◉ Assessing Technological Content of Technology Transfers in Developing Countries
- ◉ Approach 1: Sector
 - FDI: IPR -> Expansions in Chemical, Service, & Information Industries. Not in electronics & computers
 - Merchandise Imports: IPR -> Pharmaceuticals, chemicals, office & telecom, precision equipment
 - Service Imports: IPR -> Communication & computers, Licensing of Intangible Assets
- ◉ Approach 2: Impact on Local R&D, Resident Patenting, and Non-Resident Patenting
 - Foreign technologies as inputs into innovation
 - Foreign technology owners filing for patent protection

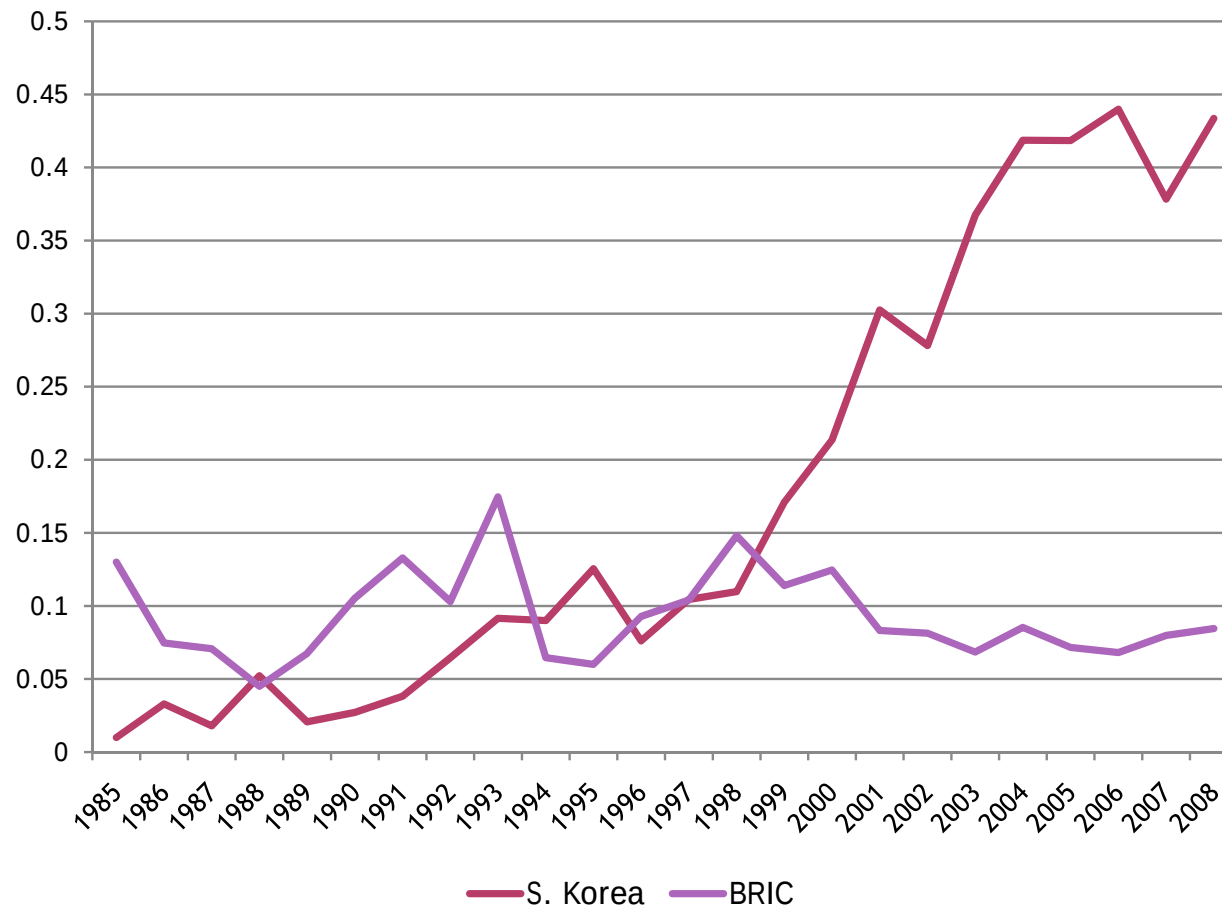
CASE STUDIES

- ◎ Brazil, Russian, India, & China (BRIC)
 - IP Developments
 - Inward Technology Transfer
 - Local Innovation & Joint Research Ventures
- ◎ South Korea
 - Source of outward technology transfers

RATIO of Outward FDI to Inward FDI Stock



RATIO of Licensing Receipts to Licensing Payments (Balance of Payments)



SUMMARY

◉ Technology Transfers

- Important to distinguish between overall level and composition (i.e. substitution effects)

◉ IPR (patent protection)

- Is one determinant of technology transfer, among others
- IP effects on Technology transfer vary by sector, level of development, imitative capacity, absorptive capacity

◉ Policy Relevance

- IPRs have potential to influence technological content of technology transfers
- Inward technology transfers can provide innovation inputs